

Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot

ID:G05-004



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INTRODUCTION

The nuclear sector has progressively expanded due to its efficiency in power generation, industrial applications, and medical technologies. However, with its benefits arise challenges, particularly the management of radioactive waste and the prevention of accidental releases. Fires in waste storage facilities present a high-risk scenario, where the atmospheric dispersion of radionuclides can endanger public health and ecosystems.

To anticipate such events, computational simulation becomes an essential strategy. This study investigates the atmospheric dispersion of radionuclides using HotSpot software to model a hypothetical fire in a radioactive waste repository. The objective is to analyze the Total Effective Dose (TED) under different meteorological and terrain conditions, supporting the development of protective measures and emergency planning in accordance with Brazilian CNEN regulations and international radiological safety standards.

MATERIALS AND METHODS

The study considered a fire scenario in a surface-level radioactive waste storage facility, leading to the release of material into the atmosphere and the formation of a dispersion plume. Such simulations support emergency planning and radiological protection assessments.

The simulations were performed using the **HotSpot** software, applying simplified terrain models (urban and rural) but with realistic configurations close to populated areas. The objective was to evaluate the influence of atmospheric and geographical factors on plume behavior and dose distribution to workers and the public.

Key parameters included a mixture of 22 radionuclides (e.g., Cs-137, Co-60, I-131, Am-241, U-238), a fire height of 7 m, an effective plume release height of 20 m, and six atmospheric stability classes (A–F, Pasquill-Gifford). Urban and rural terrains, different wind speeds, and isodose contours of 1, 10, and 20 mSv were considered, enabling a conservative assessment of radiological impacts under varied environmental conditions.

Table 1 – Meteorological Conditions Used

Wind Speed (m/s)	Meteorological Conditions	Stability Class
2.0	High sun in the sky	A – Very Unstable
3.0	High sun in the sky	B – Unstable
3.0	Low sun or overcast sky	C – Slightly Unstable
4.0	Low sun or overcast sky	D – Neutral
3.0	Nighttime	E – Slightly Stable
1.0	Nighttime	F – Very Stable

RESULTS

• Analysis of Results for Total Effective Dose (TED)

Simulations with HotSpot showed that the Total Effective Dose (TED) decreases with distance from the release point in all scenarios. Urban terrain generally resulted in lower dose values compared to rural terrain, but public dose limits (1 mSv) were reached at shorter distances in urban areas, indicating greater vulnerability near populated zones.

Atmospheric stability strongly influenced plume behavior. Unstable conditions (A–C), with strong winds and solar radiation, promoted wider dispersion and lower concentrations near the source. Stable conditions (E–F), with light winds and nighttime atmosphere, caused radionuclide accumulation close to the release point, leading to higher TED values. In class F, the plume extended less, but local doses were significantly higher.

Table 2 – Consolidated Results of TED Simulation for Radioactive Plume Dispersion across Six Meteorological Stability Classes in Urban and Rural Terrains

Stability Class	Urban Terrain – Distance to Dose Limit (km)	Rural Terrain – Distance to Dose Limit (km)
	20 mSv	1 mSv
A	0.13	0.67
B	0.094	0.55
C	0.13	0.83
D	0.14	1.10
E	0.31	3.20
F	0.69	7.00

• Analysis of Isodose Curves

Analysis of isodose curves (1, 10, and 20 mSv) confirmed that rural terrains had larger dispersion areas due to the absence of obstacles, while urban structures reduced plume spread but increased exposure risks. These findings emphasize the combined effect of meteorology, terrain, and population density on radiological risk in waste fire scenarios.

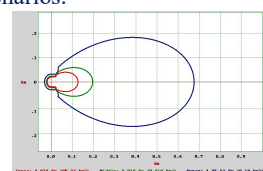


Figure 1: Isodose curves for stability class A, urban terrain.

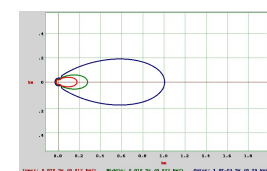


Figure 2: Isodose curves for stability class A, rural terrain.

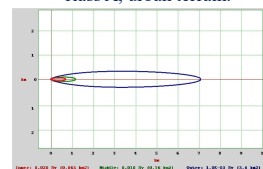


Figure 3: Isodose curves for stability class F, urban terrain.

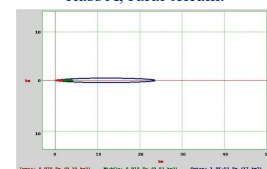


Figure 4: Isodose curves for stability class F, rural terrain.

CONCLUSIONS

The study showed that meteorological, topographical, and demographic factors influence plume dispersion and radiological risk. Unstable atmospheres increase spread, while stable conditions concentrate radionuclides near the source. Urban areas raise risk due to population density, whereas rural areas favor dilution and safer waste storage. Gaussian model simulations prove useful for emergency planning, nuclear safety policies, and public and environmental protection.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the Postgraduate Program of the Nuclear Engineering Institute (PPG-IEN) for the opportunity and support provided for the development of this research.